

WCS — AI Chat Documentation

Redacted Version — Internal Use

This is a redacted version of the internal documentation. Login credentials, staff phone numbers, individual staff names, customer phone numbers used as examples, and screenshots have been removed or replaced with generic placeholders. The workflow logic, business rules, and prompt design have been preserved for reference.

Overview

Field	Value
Document	WCS – AI Chat
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Objective

Operate an AI-assisted WhatsApp concierge for a motorcycle dealership that ingests new WhatsApp messages from a staging "Inbox Buffer," identifies whether the sender is a new or returning customer, reconstructs recent chat history, classifies the user's intent with an LLM, and then routes the request down one of three branches — pricing lookup, FAQ classification, or specs browsing — before sending a language-matched reply via the messaging provider, logging the conversation to the database, and running time-boxed follow-ups with automated human escalation when needed.

Scope

1. Communication Channel

- Automates two-way messaging on WhatsApp via a messaging provider integration.
- Provides real-time responses in both Bahasa Malaysia and English, depending on user input.

2. Data Management

Uses a structured database as the central store, organized into the following logical tables:

- Inbox Buffer — queue for unprocessed incoming messages.
- Customers — new and existing customer records.
- Chat Logs — history of past interactions, linked to customers.
- Model Pricing — structured pricing tables for vehicles/services.
- FAQ List — categorized FAQs to support automated answers.
- Staff Directory — contacts for escalation.

3. AI & Automation

- Intent Detection: classifies messages into pricing inquiry, FAQ query, or browsing/specs request.
- AI Response Generation: uses an LLM (via OpenRouter) to craft natural language replies based on context and retrieved data.

- Browsing Support: uses a live web-search API for specification or product-detail lookups.
- Prompt Engineering: tailors prompts per branch (pricing / FAQ / browse) using structured JSON and chat history.

4. Business Logic

- Customer Identification: detects whether the sender is a new or returning customer, and creates or updates records accordingly.
- Conversation Continuity: reconstructs chat history for context in replies.
- Automated Follow-ups: time-based follow-up messages (1-hour and 24-hour intervals) if there is no user response.
- Human Escalation: auto-assigns a staff member when a conversation is flagged as needing human help, or after follow-ups fail, and sends a conversation summary via WhatsApp.

5. Operational Outcomes

- Faster and more consistent replies to customer queries.
- Reduced manual workload for staff by automating common questions.
- Improved lead management through structured logs and data.
- Seamless handover to staff when automation reaches its limits.

Overall Workflow Architecture

The end-to-end flow proceeds through the following stages:

1. User sends a WhatsApp message.
2. The message is written to the Inbox Buffer staging table.
3. A scheduler/webhook in n8n picks up the next unprocessed item.
4. Context is built: chat history is reconstructed and the customer record is looked up.
5. Intent routing classifies the message as a pricing, FAQ, or browsing request.
6. Pricing branch: fetch model pricing from the database, compose a prompt, and call the LLM.
7. FAQ branch: classify the FAQ category, filter matching FAQ entries, compose a prompt, and call the LLM.
8. Browse branch: run a live web search, summarize results, compose a prompt, and call the LLM.
9. A reply is generated via the LLM.
10. The reply is sent back to the WhatsApp user via the messaging provider.
11. The exchange is logged and the Inbox Buffer record is updated.
12. A follow-up timer checks for a user reply.
13. If there is no reply, a gentle follow-up message is sent.
14. If human help is needed, a staff member is selected and notified via WhatsApp.

Database Structure

The database is organized into a small number of logical groupings:

- Model Pricing — structured pricing tables for vehicle models, colors, cash price, loan price, deposit percentages, and monthly installment breakdowns.
- AI WhatsApp Logs — contains the Customers, Chat logs, and Inbox Buffer tables.

- FAQ List — categorized FAQ entries with tags, approved answers, language, tone style, and category.
- Staff CRM — a Staff Directory table used for escalation; the workflow randomly selects any staff member marked "Active" to receive an escalation notification.

Inbox Buffer table: stores each incoming message record (message ID, sender, recipient, message body, raw payload) and a "Processed" flag that is set once the workflow has picked it up and run it through the pipeline.

Chat logs table: stores every conversation turn per customer, including a timestamp, the detected intent, the user's message, and the bot's reply.

Customers table: one record per phone number, storing the customer's name, the date the record was created, and a link to their associated chat logs.

Model Pricing table: stores per-model, per-color pricing data including cash price, loan price percentage options, deposit amounts, and monthly installment figures across multiple loan tenures.

FAQ List table: stores each FAQ's tag, the customer-facing question phrasing, the approved answer, the response language, a tone-style label, and a category (e.g., Finance, Delivery, Loan).

Staff Directory table: stores each staff member's contact identifier and an Active/Inactive status flag; only Active staff are eligible for random escalation assignment. Specific staff names and phone numbers are omitted from this document.

Access & Credentials

Login credentials for the database (Airtable) and any other connected services are managed separately and are not included in this document. Please contact your system administrator for access.

n8n Workflow — Node Overview

High-level pipeline: a WhatsApp message arrives via the messaging provider, is written to a buffer server and stored in the database, then picked up by the n8n flow on a 20-second schedule trigger, which checks for unprocessed messages.

Main pipeline nodes (functional summary):

- Customer lookup and creation — determines new vs. existing customer and creates a record if needed.
- Chat history retrieval and formatting — loads prior chat logs and assembles a context string for the current customer.
- Intent detection — an LLM call classifies the incoming message as a browse request, price inquiry, or other/FAQ request.
- Switch/routing — branches execution based on the detected intent.

Pricing Branch

A dedicated node formats model, year, and deposit details extracted from the customer's message. Pricing details are fetched from the Model Pricing table, a final prompt is built, and the LLM generates the reply.

Browse Branch

This sub-flow searches for specification or product information via a live web-search API, summarizes the results, and feeds them into the final reply prompt.

FAQ ("Other") Branch

This sub-flow classifies the FAQ category, filters and parses matching FAQ entries from the FAQ List table, combines the filtered data, and passes it into the final reply prompt.

Final Reply Generation

A "Build Final Prompt" node assembles the complete system prompt (persona, rules, business logic, and retrieved context) and calls the LLM to generate the customer-facing reply.

AI Persona & Prompt Logic

The AI persona, "Jasmin," is presented as an AI sales assistant for the dealership. Key behavioural rules encoded in the prompt:

Language Rule

Always respond in the same language the customer used (Bahasa Malaysia, English, or Chinese); the bot must not switch languages mid-conversation.

Context Provided to the Model

- Chat history for continuity.
- Customer info: name, current message, detected intent, and loan intent.
- Model pricing and color data — the model must strictly follow these values and never alter them.
- A fallback instruction for when the asked model or deposit option isn't in the price list.
- Matching FAQ info: tag, category, and tone style, plus the retrieved FAQ entries.

Conversation Flow Rules

- If the customer asks about motor models or pricing, provide the details, then ask once per conversation whether they want to proceed with cash or a loan.
- If the customer says "Cash": ask which model, give only the cash price and available colors, and ask if they'd like to proceed. If yes, reply with one fixed, language-matched confirmation phrase, unmodified.
- If the customer says "Loan": ask for the model and preferred deposit percentage (0%, 10%, 20%), then show only the monthly installment breakdown and available colors. Do not mention or send the loan application link at this stage, and do not use Markdown link formatting.
- If the customer clearly says they want to proceed with the loan application, send the loan application link exactly as configured, in the customer's language, without rewording or Markdown formatting.
- If asked whether customers from Singapore can apply: say no, unless it's a cash purchase, in which case cash-only proceeding is explained.
- If the customer asks for available models, show the model list clearly, then always guide them step by step: models first, then cash/loan pricing.

Topics Requiring Fallback (Do Not Answer Directly)

The bot must not attempt to answer the following categories, and instead responds with one fixed, language-matched fallback line indicating a human team member will follow up shortly:

- Status of loan form or approval.
- Requests to resubmit or reapply a form.
- Complaints, refunds, or delay issues.

- Discounts or price negotiation.
- Staff contact details, phone numbers, or names.
- Delivery to East Malaysia.
- Modifications or accessories not listed in the database.

General Response Guidelines

- Keep replies short and specific; answer only what the user asked.
- Do not add extra info like promotions or free gifts unless requested.
- Be friendly and professional — never overly long or "salesy."

Note: The available-models list referenced by the prompt is maintained directly in the workflow/database and is business (non-sensitive) catalog data; it has been omitted here for brevity but is not considered confidential.

Follow-Up & Escalation Logic

An "If" node checks whether the LLM has determined that escalation is needed. If yes, the workflow proceeds to escalation (staff lookup, chat-history retrieval, conversation summary, and a WhatsApp notification to the selected staff member). If not, it proceeds to the standard follow-up timer process.

Follow-up timeline:

15. 1st follow-up: 1 hour after the last message, if no user reply.
16. 2nd follow-up: 24 hours after the last message, if still no user reply.
17. If there is still no reply after the 2nd follow-up, escalate to staff after a further 1 hour.

Troubleshooting Guide

General steps:

18. Check whether the workflow is "Inactive" in n8n — reactivate if needed.
19. Open the chat-log table in the database and locate the relevant conversation.
20. Check the bot's reply for the specific issue the client is referencing.
21. If the issue relates to model price, check the price against the Model Pricing table.
22. If the bot's reply matches the Model Pricing table, but the client still says the price is wrong, cross-check against the source pricing document to confirm whether the table itself is accurate.
23. If the source document and the database price match, but the bot's reply was wrong, delete that specific chat-log entry so it doesn't pollute future context, then investigate.
24. Manually re-run the n8n workflow to observe where the issue occurs, why, and whether it is fixable.

Redaction Notice

The following categories of information have been intentionally removed from this document: database login credentials, staff names and phone numbers from the Staff Directory, example customer phone numbers shown in screenshots, and all embedded screenshots. Please refer to the secure internal credentials store for any of the above.